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The characterization of attention resource capacity and its relationship with fluid reasoning intelligence: A multiple object tracking study

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ABSTRACT

Multiple object-tracking (MOT) paradigms have the potential to highlight attention resource capacities. However, there is a dearth in research exploring the relationship between individual differences in MOT capability and higher-level cognition, such as intelligence. Previous research has demonstrated that manipulating task demands, or the task's cognitive load, can help describe this relationship. Therefore, we assessed the relationship between performance on a 3D-MOT task at different levels of cognitive load (average speed for tracking 1, 2, 3 and 4 target objects out of 8 total objects), and fluid reasoning intelligence measured by the Wechsler Abbreviated Scale of Intelligence-2nd edition (WASI-II). Also, we compared MOT performance between intellectual styles classified as: (i) low, medium or high fluid reasoning IQ, and (ii) fluid reasoning or verbal styles. As expected, speed scores decreased as target objects increased. This trend represents a proxy for attentional resource capacity as manipulations to both speed and target objects are able to highlight individual differences in available attentional resources. Furthermore, MOT capability at high load (4-targets) was the best predictor of fluid reasoning intelligence compared to lower loads (1–3 targets), and individuals with a fluid reasoning style and/or medium-high fluid reasoning intelligence outperformed individuals with a verbal style and low fluid reasoning IQ, respectively. These results describe the underlying commonalities between fluid reasoning intelligence and attention resource capacity, extending previous findings with working memory capacity. This study demonstrates that examining MOT as a measure of attention, rather than a phenomenon, can illustrate the potential to repurpose the use of this task to characterize attentional resource capacity.

1. Introduction

The relationship between cognitive capabilities and intelligence has been explored since the inception of intelligence-based measures (Heitz, Unsworth, & Engle, 2005). Primarily, this relationship has favoured the use of working memory capacity to examine how individual differences in these capacities are associated with intelligence (Engle, Kane, & Tuholski, 1999). For example, research by Kyllonen and Christal (1990) demonstrated that individual differences in working memory capacity were highly related to general intelligence, where larger capacities were associated with higher overall intelligence. Advances in further exploring this research question suggested a robust relationship to fluid reasoning intelligence, rather than general intelligence (Conway, Kane, & Engle, 2003; Engle et al., 1999; Heitz et al., 2005). Although research has targeted working memory capacity to explain this relationship, Engle et al. (1999) and Engle (2002) argue that controlled attention is ultimately, working memory capacity. Specifically, working memory is the ability to direct and sustain attention to relevant information, while ignoring irrelevant information (Engle et al., 1999). Thus, there is a need to examine whether an attentional task can help define the relationship between cognitive capacity and fluid reasoning intelligence.

Describing the relationship between attention and intelligence has proved to be difficult as both attention and intelligence are hierarchical in nature, consisting of multiple subcomponents (Heitz et al., 2005; Schweizer, 2005; Schweizer, Zimmermann, & Koch, 2000). For example, Schweizer and Moosbrugger (2004) emphasized the importance of using sustained attention as the main predictor of intelligence. Although, they along with others report a significant relationship (Crawford, 1991; Roberts, Beh, Spilsbury, & Stankov, 1991; Schweizer et al., 2000; Schweizer & Moosbrugger, 2004; Stankov, Roberts, & Spilsbury, 1994), another similar study has found contradicting results (Rockstroh & Schweizer, 2001). Similarly, the research exploring the relationship between distributed attention and intelligence have also produced opposing findings: with those supporting this relationship (Roberts et al., 1991; Roberts, Beh, & Stankov, 1988; Stankov, 1988)

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and others failing to find support (Fogarty & Stankov, 1988; Lansman, Poltrock, & Hunt, 1983). The instability of the results stemming from the examination of a single sub-component of attention has shifted the focus to a single measure of attention that accesses multiple sub-components of attention (Schweizer, Moosbrugger, & Goldhammer, 2005). Therefore, a single, accurate measure, comprised of multiple facets of attention, such as a Multiple Object-Tracking (MOT) task could further describe the relationship between visual attention capacity and intelligence.

MOT was introduced to cognitive science three decades ago by Pylyshyn and Storm (1988). The task involves visually tracking multiple objects (targets) moving around a space while ignoring other physically indistinguishable objects (distractors). A thorough dissection of the MOT paradigm has suggested that this task can tap into selective, distributed, and sustained domains of attention (see Scholl, 2009 for review). This task can account for these subcomponents of attention as it requires the participant to (i) selectively attend to target objects while ignoring distractor objects, (ii) distribute attention throughout multiple objects, and (iii) sustain this effort throughout the length of the trial (Pylyshyn & Storm, 1988). Originally, this paradigm was designed to demonstrate that attention can be allocated to multiple sources at the same time (Pylyshyn & Storm, 1988). Since then, research in MOT has explored the underlying mechanisms involved in visual tracking, and their absolute limits (Scholl, 2009). One area of research in MOT has focused on measuring the maximum number of target objects an individual can track (Alvarez & Franconeri, 2007; Cowan, 2001; Fougnie & Marois, 2006; Viswanathan & Mingolla, 2002). Particularly, this manipulation of target objects can be repurposed to examine the relationship to higher-level cognition rather than exploring the absolute limits of visual tracking. Moreover, this manipulation of target objects in MOT studies parallels research exploring the relationship between individual differences in working memory capacity and intelligence.

Two notable methods used in previous research to investigate this relationship are highlighted by (i) manipulating the working memory task's demands (i.e., manipulations in cognitive load) as well as (ii) examining performance on these working memory tasks between intellectual styles (Gevins & Smith, 2000). Manipulating a task's cognitive load can help to understand differences across individuals. In Engle et al.'s (1999) review, the authors suggest that indvidual differences in working memory capacity were demonstrated by studies that tested performance at both high and low levels of the task's cognitive load. Markedly, performance in conditions of high load are better able to explain scores on intelligence-based measures than conditions of low load (Alnæs et al., 2014; Engle, 2002; Engle et al., 1999). This idea is also supported by Gevins and Smith (2000), where a paradigm similar to the *n-back* was used to examine the impact task demands had on explaining the relationship between cognitive capacity and intelligence. Their study consisted of testing participants in a condition of high load; participants had to respond to images presented two images beforehand (i.e., 2-back condition), and conditions of low load; participants had to respond to an image presented when it matched the first image presented in the sequence. Their results also revealed that performance in conditions of high load were better predictors of intelligence than performance in conditions of low load.

Furthermore, examining the underlying differences between intellectual styles has also been demonstrated to be explanatory of individual differences in working memory capacity (Gevins & Smith, 2000). An individual with an intellectual style is defined as having a preference and/or bias towards either fluid reasoning intelligence or verbal intelligence. For example, an individual with a substantially greater fluid reasoning IQ score would be classified as possessing a fluid reasoning intellectual style. Whereas, an individual with a substantially greater verbal IQ score would be classified as having a verbal intellectual style (Gevins & Smith, 2000). Gevins and Smith (2000) classified participants into a *verbal group* if an individual's score on the verbal component of the Wechsler Adult Intelligence Scale - Revised was significantly greater than the nonverbal subscale score, and vice-versa. There were no significant differences on the working memory task between the nonverbal and verbal groups. However, their null-finding may be attributed to the high intelligence scores in the sample, where the average IQ was 121. Additionally, the researchers divided participants into three groups based on their intelligence scores as either: low, medium or high, which was relative to the sample. Their results revealed that the high- and medium-IQ groups were better at the working memory task compared to the low-IQ group. Although this finding supports the notion that working memory capacity is related to intelligence, the large memory component in the modified *n-back* task can be problematic in validating this relationship (Ackerman, Beier, & Boyle, 2005).

The condition of low load in Gevins and Smith's (2000) modified *n-back* task can be interpreted as a test of memory rather than working memory. This is problematic as memory has been found to bias the relationship between intelligence and working memory capacity due to the similarity in the construction of intelligence-based measures and memory-based measures of working memory (see Ackerman et al., 2005). This questions whether the underlying mechanisms in the cognitive capacities were properly identified, and if the measures of working memory capacity and intelligence were biased by memory (Ackerman et al., 2005).

Borrowing this methodology from research exploring individual differences in working memory capacity can help describe individual differences in attentional capacity via intelligence. MOT paradigms can provide an unbiased measure of attention as manipulating the task's cognitive load does not change the nature of the task across conditions. For instance, tracking one object is believed to access the same reserve of cognitive resources as tracking two through eight objects (Alvarez & Franconeri, 2007).

Just as working memory is capacity-limited, visuo-attentional processing is also capacity-limited (Pylyshyn & Storm, 1988); therefore, task performance suffers when task demands are greater than the individual can process (Sweller, 1994). Currently, there is an ongoing debate attempting to define this attentional capacity limit, as measured by MOT (see Suchow, Fougnie, Brady, & Alvarez, 2014 for review). Defining the limits of MOT capability has been split between the slot-based theory and resource-based theory (Suchow et al., 2014). The slot-based model posits that individuals are limited to a distinct number of target objects that they can simultaneously attend to; and thus, the task becomes impossible to successfully complete once the number of target objects is beyond the maximum number of slots available (Cowan, 2001; Luck & Vogel, 1997).

The opposing resource-based view claims that MOT capability is dependent on a pool of limited resources, which is divided across task demands, such as the number of target objects (Alvarez & Cavanagh, 2004). Therefore, if all available attentional resources are needed to track one target item at a fast speed, then only that item is captured by the individual's attention. However, if the MOT trial is *easy*, that is, items are moving at a slow speed and/or are distant from one another, then there may be leftover attentional resources available. This leftover can be assigned to additional objects or other cognitive weights associated with tracking capability. According to this theory, defining MOT performance through an item limitation alone lacks an explanation of other factors that influence the cognitive load of the visual tracking task (Alvarez & Franconeri, 2007).

The debate between the slot and resource-based theories eclipse the importance of further exploring the relationship between visual attention and separate cognitive processes (Suchow et al., 2014). This debate explains why the majority of research in this specific field explores MOT as a phenomenon, (i.e., how MOT capability is possible) rather than exploring MOT as an attention-based tool to examine the relationship of visual tracking capability with other higher-level processes, like intelligence. By exploring MOT as a paradigm, adopting the resource-based theory would be ideal in demonstrating the relationship between individual differences in attention resource capacity and fluid reasoning intelligence. For instance, research examining this capability from a slot-based perspective has failed to find a consensus in determining this object limit on the attention-based task (Suchow et al., 2014). This has resulted in a discrepancy in defining tracking capability, leading to an inconsistency in identifying an object limitation, which has ranged from four (Fougnie & Marois, 2006) to eight objects (Alvarez & Franconeri, 2007; Viswanathan & Mingolla, 2002). Moreover, it would be difficult to highlight individual differences in tracking capability from the maximum number of target items. Instead, Alvarez and Franconeri (2007) have been able to describe this limited pool of attention resources through the use of object speed as an accurate descriptor for MOT capability.

To examine MOT capability and its association to higher level processes, it may be best to use an outcome measure that accurately represents an individual's performance. Describing MOT performance by a score reflecting object speed is considered an optimal measure for visual tracking capabilities (Chen, Howe, & Holcombe, 2013; Holcombe & Chen, 2012, 2013; Störmer, Alvarez, & Cavanagh, 2014). Again, the resource-based model states that tracking multiple objects is made possible by allocating a pool of available resources to target objects (Alvarez & Cavanagh, 2004). Specifically, available resources are divided over the number of target objects (i.e., the reciprocal of the number of target objects: $1/i$; Alvarez & Franconeri, 2007). Object velocity significantly influences MOT capability (Tombu & Seiffert, 2008) and therefore, defining performance by speed would incorporate the cognitive weight speed caries when defining MOT capability relative to the deployment of attentional resources per target object. This average speed score should be represented by a function of object velocity and amount of existing resources per target set size: average speed score = speed $\times$ $(1/i)$.

Similar to Alvarez and Franconeri (2007), if we were to plot MOT capability, defined here as the average speed score at each level of cognitive difficulty, there should be a decreasing logarithmic trend. This distinct trend should describe how increasing task demands (i.e., number of target objects), impacts visual tracking capability across individuals. Ultimately, this average speed score at different levels of cognitive load, that is manipulating the number of target objects, can aid in attributing the variance in MOT performance to individual differences (Alberti, Horowitz, Bronstad, & Bowers, 2014; Faubert & Sidebottom, 2012). Additionally, this method could further examine whether a relationship exists between attention resource capacities, defined by MOT capability, and other higher-level cognitive capabilities, such as intelligence (Oksama & Hyönä, 2004).

Differences in MOT capability have been attributed to age (Trick, Perl, & Sethi, 2005), visual impairments (Alberti et al., 2014), and developmental disorders (Brodeur, Trick, Flores, Marr, & Burack, 2013; Evers et al., 2014). Despite the overlap between attention and general intelligence (Schweizer & Moosbrugger, 2004), the relationship between MOT capability and intelligence remains unclear. Counter to their initial hypothesis, Oksama and Hyönä (2004) found that MOT capability was associated with scores on the Raven's Advanced Progressive Matrices (RAPM; Raven, Court, Raven, & Kratzmeier, 1994), a measure of fluid intelligence. However, this relationship was suppressed when included in a model with other measures of visuo-spatial processing. The task used in that study was limited by its low level of difficulty, all objects were physically distinct easing object identification and discrimination. Therefore, the association between MOT and intelligence requires further exploration prior to discrediting the link between them (Oksama & Hyönä, 2004).

The current study introduces a Three-Dimensional Multiple Object-Tracking (3D-MOT) task to characterize the individual differences in MOT capability and explore its relationship to fluid reasoning intelligence. Originally designed as a two-dimensional task, recent studies have incorporated a three-dimensional component to MOT to assess the allocation of attention to multiple objects (Legault, Troje, & Faubert, 2012; Liu et al., 2005; Rehman, Kihara, Matsumoto, & Ohtsuka, 2015; Viswanathan & Mingolla, 2002). The 3D-MOT task used in this study defines performance as the average speed the participant can successfully track all target objects. The task is a better representation of a participant's capability by adapting the speed of all objects based on their performance on a trial-by-trial basis. Specifically, trial speed increases after a correct response or decreases after an incorrect response. The calculated adjustment of speed to the participant's capability can be interpreted as a better representation of the MOT capability compared to manually adjusting speed (Alvarez & Franconeri, 2007).

The purpose of the study was to characterize object-tracking capability with 3D-MOT at each level of cognitive load (i.e., the number of target objects). This method should enrich the characterization of distinct attention-based resource capacities across individuals (Faubert & Sidebottom, 2012). Therefore, it was expected that the average speed at which participants can successfully track target objects will help describe the deployment of attentional resources across multiple objects, eliciting a decreasing trend in average speed scores (available resources spent) as target objects (task demands) increase. This characterization of tracking capability can be used as a proxy for attention resource capacities, which would help describe the relationship to fluid reasoning intelligence, specifically. It was hypothesized that individuals that perform well at high levels of task load (tracking four out of eight objects) will score high on a measure of fluid reasoning intelligence. Moreover, we hypothesized that there would be no relationship with verbal intelligence. As previous research has demonstrated testing performance in conditions of high load should be associated with a larger resource capacity as there are greater task demands required for successfully completing this condition (Engle, 2002; Engle et al., 1999). Therefore, we hypothesized that 3D-MOT capability at high load should be a better predictor of fluid reasoning intelligence than conditions with lower load. These findings would contribute to if not reignite the scarcely discussed idea that individual differences in 3D-MOT capability is related to a measure of fluid reasoning intelligence.

Additionally, this paper aims to differentiate attention resource capacity, via 3D-MOT capability, across intellectual styles. Similar to Gevins and Smith (2000), we set out to explore whether 3D-MOT capability differed across two sets of intellectual styles. First, we examined whether dividing our sample into low, medium or high fluid reasoning IQ groups would demonstrate differences in 3D-MOT capability. Gevins and Smith (2000) demonstrated differences across groups with low, medium and high-IQ despite their sample's unusually high average IQ, we hypothesized that there will be differences in 3D-MOT capability across these intellectual styles, in a sample unbiased by an atypically high average IQ. However, these differences will be specific to a sample with an average IQ closer to the population average. This result would further support the link between cognitive capacity and fluid reasoning intelligence, and extend previous results favoring working memory capacity. Additionally, this question will demonstrate where significant differences in attentional resource capacities lie across fluid reasoning IQ. Secondly, we questioned whether individuals categorized by a greater score in fluid reasoning intelligence compared to verbal intelligence, and vice-versa, would perform differently on the 3D-MOT task. We hypothesized that individuals with a fluid reasoning intellectual style will outperform participants with a verbal style. A difference between these intellectual styles would suggest that individuals with a fluid reasoning intellectual style would be able to allocate more resources to task demands compared to individuals with a verbal style.

2. Methods

2.1. Participants

Seventy adult participants (35 male), between the ages of 19 to 30 years (M = 23.42, SD = 3.09), were recruited through the Montreal

Table 1
represents descriptive statistics and performance on all measures.

Measures	Mean (SD)	Range
Full Scale IQ	105.40 (14.53)	67–136
Verbal Comprehension Index (VCI)	104.90 (15.14)	67–136
Perceptual Reasoning Index (PRI)	104.43 (15.70)	67–137
Block Design *t*-score	52.59 (10.57)	28–81
Matrix Reasoning *t*-score	52.56 (10.38)	22–73
1-target avg. speed (cm/s)	279.82 (6.13)	162.55–391.00
2-targets avg. speed (cm/s)	153.80 (4.86)	64.60–292.40
3-targets avg. speed (cm/s)	114.63 (3.99)	47.55–234.94
4-targets avg. speed (cm/s)	78.84 (3.40)	29.24–167.96

Note. Means and (standard deviations) are reported. The WASI-II has an overall quotient represented by the full-scale intelligence quotient. The intelligence test is divided into two subscale scores: Verbal Comprehension Index (VCI) and Perceptual Reasoning Index (PRI). Two subtest scores in the PRI are reported: Block Design *t*-score and Matrix Reasoning *t*-score. Performance on 3D-MOT defined as the average speed score are reported across all conditions of load (1–4 target objects).

community via online classifieds. The study excluded anyone who (a) was taking medication for a pre-existing condition that would affect their attention (i.e., stimulants or sedatives), (b) had a diagnosis of ADHD, (c) had a personal or family history of seizure disorders, or (d) had any condition which would affect their vision. The McGill University Research Ethics Board Office approved the study. Furthermore, the experiment was conducted under the understanding and written consent of each participant. Table 1 presents descriptive statistics and performance on all measures.

2.2. Measures

2.2.1. Three-Dimensional Multiple Object-Tracking (3D-MOT) task

The task was controlled by a laptop and presented in 3D using a Sony HMZ-T2 Wearable Head-mounted display (HMD) with a 1280 × 720 display resolution and a field of view of 45 degrees. The virtual size of the spheres was between 20 and 55 mm (larger when they were in front of the virtual cube) and followed a linear trajectory in the 3D virtual space. Participants sat in a chair and wore the HMD. Each trial was broken down into four segments (Fig. 1). Trials began with a presentation of eight spheres positioned throughout the 3D visual field (Fig. 1a). Target spheres were highlighted as they changed from yellow to orange (four spheres shown in Fig. 1b). The spheres returned to their original color (yellow) and moved haphazardly throughout the 3D space (Fig. 1c). During this movement, the spheres could have collided with each other, altering their path. After eight seconds of movement, the spheres stopped and numbers appeared on each object to allow for identification. Participants were asked to identify all target spheres to successfully complete the trial (Fig. 1d). Subsequent task speed was generated using a 1-up 1-down staircase procedure (Levitt, 2005). Dependent on successful and unsuccessful trials, the speed of the spheres increased or decreased respectively. Initial speed was set at 68 cm per second (cm/s) displacement speed in virtual space, depending on previous trial performance, object speed increased or decreased by 0.05 log. Possible speeds ranged from 0.06 cm/s to 544 cm/s.

2.2.2. Wechsler Abbreviated Scale of Intelligence – second edition (WASI-II)

Participants also had their cognitive ability assessed by the WASI-II Full Scale IQ-4 Subtests (FSIQ-4) comprised of the Perceptual Reasoning Index (PRI) and Verbal Comprehension Index (VCI) subscales. This measurement of general intelligence between the ages of 6 and 89 years old is made up of four subtests: Block Design (BD), Vocabulary (VC), Matrix Reasoning (MR), and Similarities (SI) represented by standardized age- and gender-based norms, on a *t*-distribution (Wechsler, 2011).

2.3. Procedure

Participants completed the 3D-MOT task and then were assessed on the WAS-II. The study took between 60 and 90 min to complete, depending on the participant's performance.

In the 3D-MOT task, all participants were asked to track one, two, three, and four targets out of a total of eight objects in separate blocks. There was no set number of trials per condition, instead the block ended once six inversions occurred. An inversion signified a shift from a wrong answer to a right answer or vice versa. After six inversions, an average speed score was computed. Performance was defined by the geometric mean of trial speed at all six inversions. The geometric mean was used because the scores at each inversion produce a product rather than a summed total. That being, trial speed increases or decreases proportionally rather than linearly: object speed increased or decreased by 0.05 log. Therefore, in order to account for these differences in object velocity between trials, the geometric mean was used to calculate the average speed score. Participants completed all conditions (tracking one, two, three, and four out of eight total spheres) twice and the outcome variable for each condition was the average of both average speed scores. The order for condition of load was counterbalanced for all participants by generating a random sequence. Upon completion of the 3D-MOT task, participants completed the WASI-II.

2.4. Design and analyses

Descriptive statistics (mean, standard deviation, and range) for all participants were presented per variable of interest. First, we conducted a one-way repeated measures ANOVA to explore the differences in average speed scores across conditions of load (one, two, three, and four target objects). This analysis was conducted to validate that the use

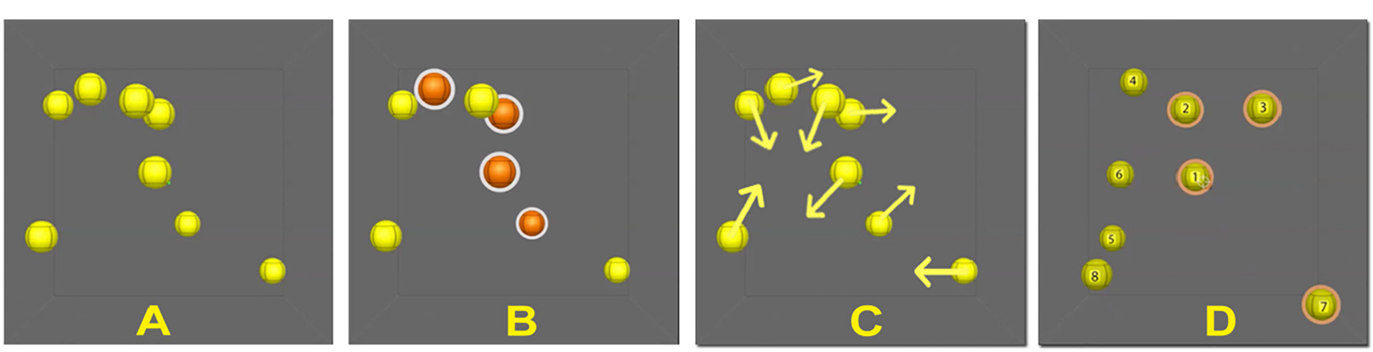

Fig. 1. A diagram representing a typical trial of the 3D-MOT task used in the present study. (A) A presentation of all eight spheres in a 3D space. (B) The desired spheres are highlighted (four presented here), indexing which objects are to be tracked. (C) The spheres then move randomly throughout the visual field for 8 s. (D) When stopped, numbers appear on all 8 spheres and the participant must identify the indexed sphere.

of an average speed score was consistent with previous findings that indicate a decreasing trend in performance as target objects increase. All a priori assumptions for a repeated measures ANOVA were met except for sphericity. To correct for this violation, we applied a Greenhouse-Geisser correction to the analysis. Furthermore, 3 paired-samples *t*-test analyses were conducted to determine whether the average speed scores at conditions of load significantly differed from one another. A Bonferroni correction was applied to control for multiple comparisons.

Next, we conducted five multiple regression analyses to explore the relationship between 3D-MOT capability at different levels of load on verbal and fluid reasoning intelligence, as measured by the WASI-II. The first two multiple regression analyses used 3D-MOT capability at high load (composite score of average speed at 3 and 4 targets) and low load (composite score of average speed at 1 and 2 targets) to predict performance on (i) the Perceptual Reasoning Index (PRI) subscale score and (ii) the Verbal Comprehension Index (VCI) subscale score. The next three multiple regression analyses used 3D-MOT performance at 3 and 4 target items as predictors to account for: (i) performance on the PRI subscale score, (ii) performance on the block design task (t-score), and (iii) performance on the matrix reasoning task (t-score). All a priori assumptions were met for the four multiple regression analyses.

We then examined the differences in 3D-MOT capability between intellectual styles: (i) low vs. medium vs. high and (ii) fluid reasoning versus verbal intelligence style. We divided participants into three groups based on their scores on the measure of fluid reasoning intelligence on the WASI-II PRI subscale. We conducted two one-way between subjects ANOVAs examining the differences in 3D-MOT capability in conditions of (i) high load, which was defined by the composite of the average speed scores for 3 and 4 targets; and (ii) low load, which was defined by the composite of the average speed scores for 1 and 2 targets, by PRI level (Low Fluid Reasoning IQ [LFRIQ], Medium Fluid Reasoning IQ [MFRIQ], and High Fluid Reasoning IQ [HFRIQ]). All a priori assumptions were met for both ANOVAs.

Lastly, we conducted two one-way between subjects ANOVAs exploring differences between fluid reasoning and verbal intellectual styles at high- and low load. Specifically, we conducted a one-way between subjects ANOVA with the composite of the average speed scores for 3 and 4 targets between individuals defined by a fluid reasoning intelligence style versus participants defined by a verbal intelligence style. Similarly, we conducted a one-way between subjects ANOVA with the composite of the average speed scores for 1 and 2 targets between participants defined individuals defined by a fluid reasoning intelligence style versus individuals defined by a verbal intelligence style. All a priori assumptions were met for both ANOVAs.

3. Results

3.1. Cognitive load and attention resource capacity

As expected, the repeated measures ANOVA with Greenhouse-Geisser correction revealed a decreasing trend for speed scores as the number of target objects increased, $F(2.16, 148.81) = 726.96$, $p < .001$, partial $\eta^2 = 0.91$. The decreasing trend is represented in Fig. 2a: 1 vs. 2, $t(69) = 23.28$ $p < .001$; 2 vs. 3, $t(69) = 9.59$, $p < .001$; 3 vs 4, $t(69) = 13.58$, $p < .001$ (with a corrected Bonferroni value of 0.05/3 = 0.017). Like Alvarez and Franconeri (2007) the decreasing trend describing 3D-MOT performance at increasing levels of load mapped onto a logarithmic equation, defined here as: $y = 144.3\ln(x) + 271.38$ at $R^2 = 0.981$ (see Fig. 2b).

To characterize the trend of 3D-MOT performance at increasing levels of task difficulty, the log average speed scores were plotted to the log of the number of target objects (log-log plot), resulting in a strong linear correlation ($r^2 = 0.991$; see Fig. 3). This declining trend in average speed scores as task difficulty increased across conditions resembled a logarithmic function (see Fig. 2b).

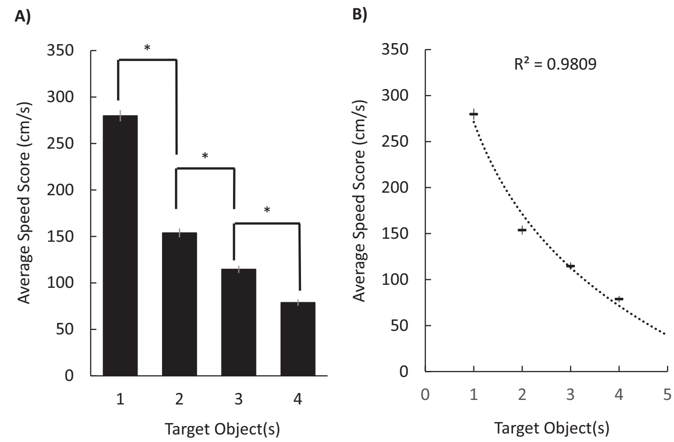

Fig. 2. A) A representation of 3D-MOT average speed scores by load conditions. Average speed scores across conditions of load, the number of target objects. The bar graph represents scores when tracking: 1-object ($M = 279.82$, $SE = 6.13$), 2-objects ($M = 172.35$, $SE = 11.61$), 3-objects ($M = 130.66$, $SE = 8.45$), and 4-objects ($M = 91.83$, $SE = 8.12$). * $p < .001$. B) A graph plotting the average speed scores by target objects (1–4). 3D-MOT capability as target objects increase is defined by the logarithmic equation: $y = 144.3\ln(x) + 271.38$ at $R^2 = 0.981$.

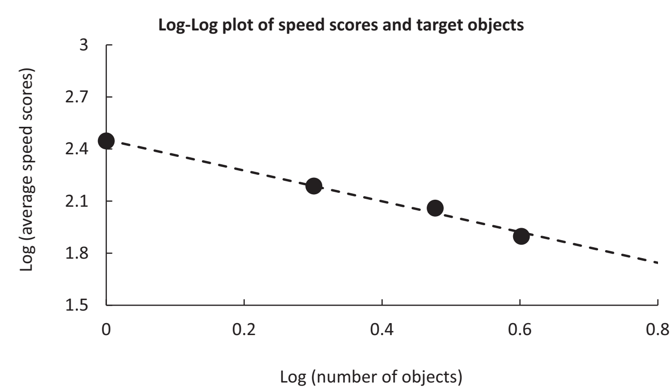

Fig. 3. The log of average speed scores (performance) were plotted onto the log of target objects (condition). The points on the log-log mapped onto a linear function:
$y = -0.8858\times + 2.4531$ at $r^2 = 0.991$.

Again, the *flexible-resource model* states that available resources are divided among the task conditions, in this case the number of target objects. According to this model, an individual is able to allocate $\left(1/i\right)$ resources, where i represents the number of target objects in an 3D-MOT trial. The linearity of plotted points on the log-log plot, or the decreasing logarithmic trend of speed by target objects, suggests that the velocity at which target objects are moving is proportional to the amount of attentional resources required for successfully completing the tracking trial. Therefore, the function $y = s \times 1/i$, where (y) average speed score is a function of speed (s) and resources required $\left(1/i\right)$ to successfully track i objects can be used to describe 3D-MOT performance.

3.2. Intelligence and 3D-MOT capability by load

A multiple regression analysis was conducted to explore the relationship between low and high conditions of load on 3D-MOT and the Perceptual Reasoning Index subscale score (PRI; i.e., fluid reasoning intelligence). The composite score of average speed scores at 1 and 2 target objects (low load) and the composite score of average speed scores at 3 and 4 target objects (high load) were used as predictors for

Table 2
Standard multiple regression examining the effect of conditions of high and low load on the perceptual reasoning index and verbal comprehension index of the WASI-II.

Predictors	B	SE (B)	Beta	Pearson R	sr^2	p
Perceptual reasoning index: $R^2 = 0.17$, Adjusted $R^2 = 0.15$.						
Constant	91.27	9.53				$p < .001$
Low Load (1 & 2 target objects)	−0.03	0.03	−0.15	0.17	0.01	$p = .235$
High Load (3 & 4 target objects)	0.14	0.04	0.50	0.40	0.14	$p < .001$
Verbal Comprehension Index: $R^2 = 0.0.02$, Adjusted $R^2 = -0.02$.						
Constant	115.46	9.57				$p < .001$
Low Load (1 & 2 target objects)	−0.02	0.03	−0.11	−0.12	0	$p = .524$
High Load (3 & 4 target objects)	0	0.04	0.02	0.09	0	$p = .918$

Note. sr^2 denotes the squared semi-partial correlation.

PRI scores. The model was statistically detectable: $F(2, 67) = 7.01$, $p = .002$, $R^2 = 0.17$, Adjusted $R^2 = 0.15$; however, variance in PRI scores was accounted for by 3D-MOT capability at high load: $b = 0.14$, $t(67) = 3.41$, $p = .001$, 95% CI [0.054, 0.219], but not low load: $b = -0.03$, $t(67) = -1.06$, $p = .295$, 95% CI [−0.084, 0.029]. This analysis demonstrates that performance at higher load accounted for a significant amount of the variance in PRI scores. Next, we examined whether low and high conditions of load on 3D-MOT could predict Verbal Comprehension Index subscale scores (VCI; i.e., verbal intelligence). Five participants were removed from this analysis since English was not their first language and were therefore unable to complete the VCI subtests. The model was not statistically detectable: $F(2, 62) = 0.46$, $p = .632$, $R^2 = 0.02$, Adjusted $R^2 = -0.02$ with 3D-MOT capability at high load: $b = 0$, $t(62) = -0.10$, $p = .918$, 95% CI [−0.08, 0.04] and low load: $b = -0.02$, $t(62) = -0.64$, $p = .524$, 95% CI [−0.08, 0.04] (see Table 2).

The relationship between 3D-MOT capability in high load conditions, 3D-MOT performance at 3 and 4 target objects, were associated with PRI: $F(2, 67) = 6.87$, $p = .002$, $R^2 = 0.17$, Adjusted $R^2 = 0.15$. Only 3D-MOT performance at 4 target objects was significant: $b = 0.19$, $t(67) = -2.05$, $p = .045$, 95% CI [−0.024, 0.409], while 3D-MOT performance at 3 target objects was not: $b = -0.04$, $t(67) = -0.47$, $p = .637$, 95% CI [−0.141, 0.203]. Further exploring the association between PRI and high load conditions of 3D-MOT demonstrate that once again, higher load is associated with the variance in PRI scores (see Table 3).

A fourth multiple regression exploring the association between 3D-MOT capability in high load conditions and performance on the PRI subtest Block Design *t*-score was significant: $F(2, 69) = 8.77$, $p < .001$, $R^2 = 0.207$, Adjusted $R^2 = 0.184$. As expected, performance at 4 objects was significant: $b = 0.136$, $t(67) = 2.20$, $p = .031$, 95% CI [−0.014, 0.285], while performance at 3 target objects was not significant: $b = 0.035$, $t(67) = 0.663$, $p = .510$, 95% CI [−0.092, 0.163]. Again, there was a significant association at high load (4-targets) and the Block Design subtest of the PRI subscale (see Table 3).

Finally, a fifth multiple regression explored the association between 3D-MOT capability in high load conditions and the PRI subtest Matrix Reasoning *t*-score. The model was not significant, $F(2, 67) = 2.45$, $p = .094$, $R^2 = 0.068$, Adjusted $R^2 = 0.040$. at either 3 or 4 target objects. Counter to previous findings Oksama and Hyönä (2004), there was no association with performance at high loads on 3D-MOT and the Matrix Reasoning subtest (see Table 3).

Next, we explored where differences in MOT performance, or attentional resource capacity lie across fluid reasoning intelligence. Participants were divided into one of three groups based on their performance on the fluid reasoning subscale score of the WASI-II (PRI subscale), given the aforementioned link with working memory capacity (Conway et al., 2003; Engle et al., 1999; Heitz et al., 2005). Similar to Gevins and Smith (2000), participants were categorized into the three groups based on their percentile rank generated from the PRI subscale score of the WASI-II. Therefore, participants were categorized as either: (i) Low fluid reasoning IQ (LFRIQ) consisting of a percentile rank from 0 to 33.32 ($n = 20$; $M = 85.55$, $SD = 10.27$; Range [67–99]), (ii) Medium fluid reasoning IQ (MFRIQ), corresponding to a percentile between 33.33 and 66.65 ($n = 27$; $M = 104.19$, $SD = 2.65$, Range [100–109]), or (iii) High fluid reasoning IQ (HFRIQ) with a percentile rank between 66.66 and 100 ($M = 121.13$, $SD = 15.70$, Range [110–137]; see Table 4). The composite score of average speed scores at 1 and 2 target objects (low load) and the composite score of average speed scores at 3 and 4 target objects (high load) are used for the subsequent analyses on intellectual styles.

A one-way between subjects ANOVA revealed a main effect of fluid reasoning IQ level, as measured by the PRI subscale score, on a composite score of 3D-MOT performance in conditions of high load: $F(2, 67) = 6.29$, $p = .003$, partial $\eta^2 = 0.16$. Post-hoc analyses revealed a significant difference between LFRIQ ($M = 157.70$, $SD = 58.78$) and HFRIQ ($M = 212.29$, $SD = 62.25$), $p = .004$ *Hedges' g* = 0.90; and LFRIQ ($M = 157.70$, $SD = 58.78$) and MFRIQ ($M = 203.92$, $SD = 41.83$): $p = .014$ *Hedges' g* = 0.93. There was no statistically detectable difference between MFRIQ ($M = 203.92$, $SD = 41.83$) and HFRIQ ($M = 212.29$, $SD = 62.25$) $p = .849$, *Hedges' g* = 0.16. Furthermore, a second one-way between subjects ANOVA was conducted to explore the differences in MOT performance in conditions of low load by low, medium, and high fluid reasoning IQ scores. There was no statistically detectable differences between low load and PRI level: $F(2, 67) = 1.32$, $p = .274$, partial $\eta^2 = 0.04$ (see Fig. 4).

Table 3
Multiple regression analyses examining the relationship between performance on conditions of 3 and 4 target objects on Perceptual Reasoning Index of the WASI-II, Block Design Task t-scores, and Matrix Reasoning Task t-scores.

Predictors	B	SE (B)	Beta	Pearson R	sr^2	P
Perceptual Reasoning Index: $R^2 = 0.17$, Adjusted $R^2 = 0.15$.						
Constant	84.94	6.27				$p < .001$
3 target objects	−0.04	0.08	0.35	0.34	0.00	$p = .637$
4 target objects	0.19	0.09	0.08	0.41	0.05	$p = .045$
Block Design Task t-scores: $R^2 = 0.21$, Adjusted $R^2 = 0.18$.						
Constant	37.86	4.12				$p < .001$
3 target objects	0.04	0.05	0.11	0.39	0.01	$p = .510$
4 target objects	0.14	0.06	0.37	0.45	0.06	$p = .031$
Matrix Reasoning Task t-scores: $R^2 = 0.07$, Adjusted $R^2 = 0.04$.						
Constant	47.57	4.39				$p < .001$
3 target objects	−0.05	0.06	−0.14	0.12	0.01	$p = .427$
4 target objects	0.13	0.07	0.35	0.24	0.05	$p = .055$

Note. sr^2 denotes the squared semi-partial correlation.

Table 4
Means and standard deviation on fluid reasoning intelligence and 3D-MOT by PRI level.

PRI Level	n	PRI	High Load	Low Load
Low Fluid Reasoning IQ (LFRIQ)	20	85.55 (10.27)	157.70 (58.78)	409.58 (83.70)
Medium Fluid Reasoning IQ (MFRIQ)	27	104.19 (2.65)	203.92 (41.83)	439.42 (79.61)
High Fluid Reasoning IQ (HFRIQ)	23	121.13 (15.70)	212.29 (62.25)	447.72 (78.40)

Note. LFRIQ classified by 20 participants with PRI scores between 0 and 33.32 percentile rank, ranging from 67 to 99 (inclusively). MFRIQ has 27 participants grouped by PRI scores between 33.33 and 66.32 percentile rank, ranging from 100 to 109 (inclusively). HFRIQ is comprised of 23 participants with PRI scores between 66.33 and 100 percentile ranks, ranging from 110 to 137.

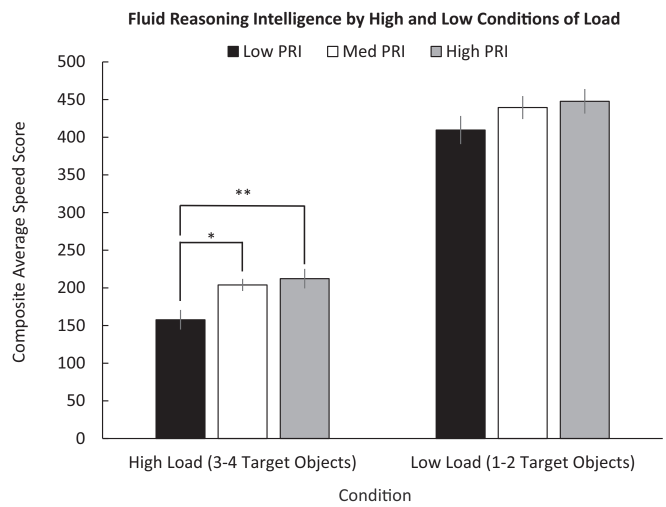

Fig. 4. A representation of performance in 3D-MOT capability in conditions of high (bars on the left) and low (bars on the right) between levels of fluid reasoning ability. Fluid reasoning ability was defined as either low (67–99; black), medium (100–109; white), and high (110–137, grey) by dividing sample into three groups across the 33rd and 66th percentiles. * $p = .014$, *Hedges g* = 0.93 ** $p = .004$, *Hedges g* = 0.90.

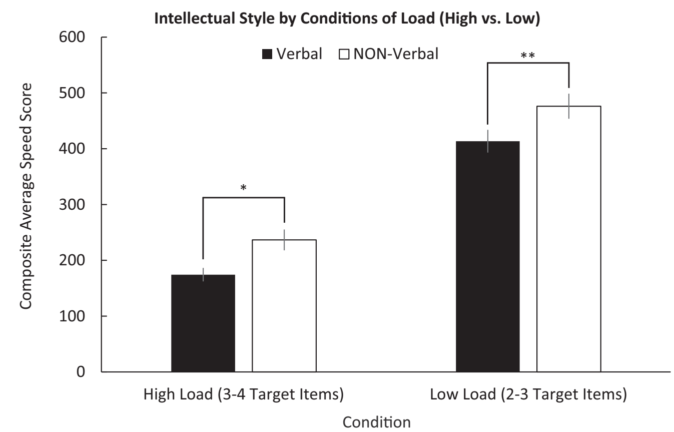

Fig. 5. A representation of performance in 3D-MOT capability in conditions of high (bars on the left) and low (bars on the right) between intellectual styles. Participants were classified in the fluid reasoning group (black) if their scores on the fluid reasoning subscale of the intelligence-based measures was significantly higher than the verbal subscale. Participants were classified in the verbal group (white) if their scores on the fluid reasoning subscale of the intelligence-based measures was significantly higher than the fluid reasoning component. * $p = .009$, partial $\eta^2 = 0.21$, ** $p = .047$, partial $\eta^2 = 0.13$.

Lastly, we explored whether there was a difference in 3D-MOT performance between intelligence styles. We borrowed the methodology used by Gevins and Smith (2000), where they examined the differences in working memory capacity as a function of intellectual styles (i.e., higher verbal intelligence than fluid reasoning intelligence and vice-versa). Our study followed a similar procedure to classify a participant as either *fluid reasoning* or *verbal* in an attempt to examine the differences between these two intellectual styles based on our proxy for attentional resource capacity. Five subjects were removed from this analysis as they did not complete the verbal section of the WASI-II because English was not their first language. First, every participant's fluid reasoning subscale score (PRI) was subtracted from the verbal subscale (VCI) score and then divided by the full-scale IQ score (FSIQ). Negative values can be interpreted as a fluid reasoning intellectual style, where values further away from zero are indicative of a stronger fluid reasoning style. Likewise, positive values can be interpreted as a verbal intellectual style, where values further away from zero are indicative of a stronger verbal style. Thus, the top 25% of these values were assigned to the verbal group ($n = 16$) while the bottom 25% were assigned to the fluid reasoning group (n = 16) to examine whether there were any differences in attentional capacity, as measured by 3D-MOT, between these groups (see Table 5).

A one-way between subjects ANOVA examined the differences in 3D-MOT performance in conditions of high load on the 3D-MOT task (i.e., average speed scores at 3 and 4 target objects) between the verbal group and fluid reasoning group. The results revealed that the fluid reasoning group ($M = 236.58$, $SD = 15.72$) outperformed the verbal group ($M = 174.33$, $SD = 15.72$) on the high load conditions: $F(1, 30) = 7.84$, $p = .009$, partial $\eta^2 = 0.21$. Similarly, a one-way between subjects ANOVA that examined the differences in 3D-MOT performance in conditions of low load (i.e., average speed scores at 1 and 2 target objects) was also significant: $F(1, 30) = 4.27$, $p = .047$, partial $\eta^2 = 0.13$. Once again, the fluid reasoning group ($M = 476.17$, $SD = 89.89$) outperformed the verbal group ($M = 413.47$, $SD = 81.48$) on conditions of low load (see Fig. 5).

Table 5
Means and standard deviation for intellectual styles on WASI-II and 3D-MOT at low and high load.

Intellectual Styles	FSIQ	VCI	PRI	High Load	Low Load
Verbal (n = 16)	107.06	116.44	95.50	174.33	413.47
	(15.98)	(15.11)	(15.07)	(48.37)	(81.48)
Fluid reasoning (n = 16)	104.38	94.56	114.62	236.58	476.17
	(11.27)	(9.29)	(12.67)	(74.63)	(89.88)

Note. Means and (standard deviations) reported for 16 participants in both verbal and fluid reasoning intellectual styles.

4. Discussion

The aim of this study was to determine whether performance on a 3D-MOT task at different levels of cognitive load can describe the relationship between MOT capability and fluid reasoning intelligence. To do so, MOT capability was defined by a continuous ratio variable, representative of task demands: the average speed score. Our findings concur with similar research conducted by Alvarez and Franconeri (2007), where the average speed that participants successfully tracked *all* target objects decreased as a function of set size. With the validation of the average speed score as a suitable metric for MOT capability and ultimately, attention resource capacity, our results revealed that visual tracking capability was positively associated with fluid reasoning intelligence. The number of target objects was manipulated to extrapolate individual differences across participants between conditions of low and high cognitive load. Here, performance in conditions of high load, (i.e., tracking four target objects out of eight total objects), strengthened the association between fluid reasoning IQ and 3D-MOT capability, and that performance on the 3D-MOT task differed between the low ability group and higher fluid reasoning IQ ability groups. Furthermore, the results also highlighted differences between fluid reasoning and verbal intellectual styles where the fluid reasoning group outperformed the verbal group on the 3D-MOT task. Overall, these results suggest that an average speed score is an appropriate metric to (i) characterize 3D-MOT capability across individuals as attention resource capacity, (ii) explore the relationship between 3D-MOT capability and fluid reasoning intelligence, and (iii) extend previous findings by Gevins and Smith (2000) by accurately manipulating an attention-based task's cognitive load to help describe differences between intellectual styles.

Previous research exploring MOT capability has been limited in their ability to define tracking capability (Suchow et al., 2014). We propose that the average speed score used in the current study is an appropriate descriptor of tracking capability. In comparison to the Alvarez and Franconeri (2007) study, where trial speed was set by the participants themselves, our average speed score used a staircase procedure to define performance. Through the use of this methodology, increasing the number of target objects (i.e., from one to four out of eight objects) resulted in a decreasing logarithmic trend in tracking capability. This representation is consistent with the decreasing logarithmic trend in Alvarez and Franconeri (2007), which was used to describe the allocation of attentional resources to task demands.

Furthermore, the decreasing trend in task performance as target objects increased are in-line with the *flexible-resource model* as evidenced by the linear relationship of the log-log plot, which demonstrates that object velocity was proportional to the amount of resources available for target objects. Objects moving at faster speeds demand additional effort or attentional resources to track (Alvarez & Franconeri, 2007; Bettencourt & Somers, 2009; Tombu & Seiffert, 2008). If available resources cannot be divided among all target objects while accounting for object velocity, then trial performance suffers (Alvarez & Franconeri, 2007). Therefore, attention is taxed by both speed and number of target objects, and objects are lost from attention or swapped for distractors when task demands are greater than available cognitive resources (Drew, Horowitz, & Vogel, 2013; Tombu & Seiffert, 2008). This finding extends previous research, which was limited by categorical speed manipulations used to define MOT capability (Drew et al., 2013; Fougnie & Marois, 2006; Franconeri, Jonathan, & Scimeca, 2010; Oksama & Hyönä, 2004; Pylyshyn & Storm, 1988; Trick, Jaspers-Fayer, & Sethi, 2005). Whereas Alvarez and Franconeri (2007) allowed participants to manually adjust trial speed to their capability, our results suggest that the average speed score, which better represents the average speed at which the participant successfully tracked all target objects, is a refined representation of attentional resource capacity. However, the use of speed to define attentional resource capacity in MOT capability contradicts previous findings suggesting that speed does not affect, nor represents tracking capability (Franconeri et al., 2010; Franconeri, Lin, Enns, Pylyshyn, & Fisher, 2008; Iordanescu, Grabowecky, & Suzuki, 2009).

The issue with using speed as an outcome variable, representative of MOT capability, may depend whether MOT is examined as a paradigm or as a phenomenon. For instance, Tombu and Seiffert (2008) isolated the effects of speed on a MOT task through the introduction of a dual-task and they found that increasing speed required additional cognitive resources to process in order to successfully complete the dual task. Conversely, Franconeri et al. (2010) claimed that MOT capability was influenced by object spacing alone, and that: “people could reliably track an unlimited number of objects as fast as they could track a single object” (p. 920). Their argument posits that objects moving at higher speeds result in greater instances of interactions between objects and the resulting object spacing constraint negatively impacts tracking capability. One prominent factor overlooked in Franconeri et al.'s (2010) study is their attempt to isolate speed by reducing tracking time. To demonstrate object proximity played a larger role than speed, the researchers held the distance objects travelled constant (i.e., the number of possible interactions) and manipulated object speed by fast-forwarding the trial at 2 times, or slowing down trial presentation by half the time, and one quarter of the time. This manipulation may better represent the costs of sustained attention, rather than an isolation of object speed. In Tombu and Seiffert's (2008) isolation of speed and object proximity, both factors resulted in greater difficulty to complete the MOT task when combined with a separate auditory task, with all other variables (e.g., tracking time) being equal. This finding demonstrates that both speed and object proximity require additional attentional resources to successfully process (Bettencourt & Somers, 2009). This discrepancy between findings suggests that using speed as a characterization of individual differences in tracking capability examines MOT capability as a measure of attention, rather than MOT as a phenomenon (Scholl, 2009). We therefore suggest that defining performance by speed can be used as an improved characterization of attentional capacity as successfully tracking objects at higher speeds results in more interactions between objects, and in turn this requires more cognitive resources to be deployed.

The validation of an average speed score as a metric to describe MOT capability, infers the possibility to explain individual differences in attentional resource capacities by higher-level cognitive functioning. Previous research has found a separation between individuals in object-tracking capability that can be highlighted by defining MOT performance across levels of cognitive load (Alnæs et al., 2014). To our knowledge, our study is the first study to manipulate the number of target objects to examine whether performance in conditions of high and/or low load in a MOT paradigm can be used to predict fluid reasoning intelligence. Our results revealed that performance at high load; specifically, tracking four out of eight target objects, was the best predictor of fluid reasoning intelligence, as measured by the Perceptual Reasoning Index subscale score (PRI), than conditions with lower load: tracking one, two, or three target objects. This finding builds on previous research by Alnæs et al. (2014), where it was determined that individual differences in MOT capability in conditions of high load were associated with increased pupil size. Larger pupil sizes exhibited while performing conditions of high load was argued to be representative of the amount of cognitive resources, or mental effort, that an individual was able to allocate to the MOT task (Alnæs et al., 2014). The current study builds on these findings by demonstrating that the average speed score can be used as a proxy to describe attentional resource capacity. We therefore suggest that speed can be used as an indicator of performance to further explore the association between MOT capability and intelligence.

The attention-based nature of the 3D-MOT task and its relationship to higher-level cognition is in line with Engle et al.'s (1999) argument that attention plays a large role in higher-level cognitive capacity. The relationship between MOT capability at low and high load was established for fluid reasoning intelligence. However, tracking capability in either condition was unable to predict verbal intelligence. These results emphasize the nonverbal nature of the MOT task. The relationship between MOT capability and fluid reasoning intelligence, but not verbal intelligence, is indicative of the similarities between the nonverbal nature of both tasks. Markedly, this could be beneficial for examining attention resource capacity in non-verbal populations, or for individuals whose language abilities are underdeveloped.

Furthermore, fluid reasoning intelligence, or the WASI-II's PRI subscale, is related to analytical reasoning (Wechsler, 2011) and working memory capacity (Oberauer, Schulze, Wilhelm, & Süß, 2005). This association may be due to the similarities in 3D-MOT properties and facets of the PRI subscale. Particularly, 3D-MOT requires incorporating target objects as a whole and parsing relevant from irrelevant information; this is related to aspects of the working memory which in turn are related to sustained attention, a significant predictor of PRI (Buehner, Krumm, Ziegler, & Pluecken, 2006). The established relationship between visual tracking capability and fluid reasoning intelligence questions the conclusions made by previous research, that has discredited the role of fluid reasoning intelligence and MOT capability (Oksama & Hyönä, 2004).

The methodological differences in our study compared to the initial examination of MOT capability and fluid reasoning IQ by Oksama and Hyönä's (2004) MOT task are highlighted by (i) the manipulation of the task's cognitive load, and (ii) the definition of performance by an average speed score. As previously discussed, these considerations in the current study expands the knowledge on the association between visual tracking capability and fluid reasoning intelligence. Furthermore, the choice of intelligence-based measure furthers the discussion between MOT capability and fluid reasoning IQ from the Oksama and Hyönä (2004) study. The two subtests that make-up the PRI subscale in the WASI-II, the block design (BD) and Matrix Reasoning (MR) tasks, are measures of fluid reasoning intelligence: examining object organization, spatial visualization, and simultaneous processing (Groth-Marnat & Teal, 2000). By further exploring the role of PRI in MOT capability, we compared *t*-scores on the two subtests (BD and MR) with 3D-MOT average speed scores at high levels of difficulty. This was done to explore the effect of suppression MR had on MOT capability in previous work (Oksama & Hyönä, 2004). The positive relationship between BD and 3D-MOT capability accounts for a large portion of the variance, while there was no significant association with MR. The BD task requires an estimation of object distance and sustained attention to complete, thus elucidating the commonalities between 3D-MOT and BD (Oberauer et al., 2005). Additionally, the use of depth in both tasks can be used to explain the association between 3D-MOT and BD as depth perception plays a vital role in attentional processing (Vidaković & Zdravković, 2010). For instance, the use of depth facilitates object discrimination (Viswanathan & Mingolla, 2002), a shared characteristic in both the BD subtest where physically manipulating and positioning the blocks to replicate an image, and in the 3D-MOT task, where tracking multiple objects during instances of object occlusion and collision. The relationship between 3D-MOT capability and fluid reasoning intelligence contradicts previous research by Oksama and Hyönä (2004) where their relationship between MOT and fluid reasoning intelligence, as measured by the RAPM was suppressed. However, this study extends the previous finding with the inclusion of another measure of fluid reasoning intelligence such as the BD task.

A relationship between attention resource capacity, implied by the average speed score, and fluid reasoning intelligence was further supported by the differences in groups divided by performance on the PRI subscale. Here, individuals in the medium and high ability groups, defined as the mid and upper thirds of the sample's PRI subscale scores, outperformed participants in the lower ability group, defined as the bottom third of the sample's PRI scores, on the 3D-MOT task in conditions of high load. This finding concurs with Gevins and Smith (2000) study that demonstrated a difference between fluid reasoning IQ ability groups on a working memory task. However, their biased sample of individuals with an average IQ of 121, may not be the best representation of the general population as their associations may be the result of a restricted range. Our sample, with an average IQ closer to the general population, highlighted the difference in attentional capacity between the higher and lower ability groups. The differences between higher (high and medium groups fluid reasoning IQ groups) and low IQ groups suggest that significant differences in attentional resource capacities lie at about an IQ of 100. Although there were no differences between the medium and high groups, their position above the population mean IQ, coupled with the linear relationship discussed earlier, demonstrates a clear difference in the allocation of attention resource capacity to task demands. Specifically, these results suggest that individuals with a greater fluid reasoning IQ than individuals with a lower IQ are able to allocate more attentional resources to the increasing cognitive demands of the 3D-MOT trial. Moreover, these results are salient across the population average. Future research could benefit from further exploring the underlying cognitive differences, specifically related to attention resource capacity, that lie across the population average. Overall, the results presented here extends Gevins and Smith's (2000) findings to a measure of attention capacity, rather than a modified *n-back* task, which was deemed a measure of working memory capacity.

Additionally, the better performance on the visual attention-based task exhibited by the group representative of a fluid reasoning intellectual style demonstrates the differences in attention resource capacity between fluid reasoning and verbal intellectual styles. Again, these results extend the work by Gevins and Smith (2000) that did not detect a difference in performance on their working memory task between these two intellectual styles. Markedly, these differences were highlighted in conditions of high load, which suggests that individuals with higher fluid reasoning intelligence, were able to allocate more attentional resources to the increasing task demands of the 3D-MOT task than the verbal intellectual style group. These findings can be explained by individual differences in the perception of task difficulty, which stems from the individual's level of cognitive competency (Paas & van Merriënboer, 1994). This link between cognitive competency and task difficulty implies that those individuals who found a trial more difficult than the next individual would have a lower attentional resource capacity. Specifically, the individual may perceive a greater degree of difficulty as they cannot allocate the appropriate amount of resources than another individual that can. This discrepancy in cognitive competency was highlighted by the association between tracking capability at challenging levels of difficulty (i.e., tracking in conditions of high load) and fluid reasoning intelligence. To our knowledge, these results are the first to demonstrate a link between attention resource capacity and individuals characterized by a fluid reasoning intelligence style.

Although this study demonstrated a link between the fluid reasoning subscale score of the WASI-II, a possible limitation can be seen with the accuracy of the abbreviated IQ measure. We encourage future research to examine the relationship between MOT capability and multiple measures of fluid reasoning intelligence made available by the Wechsler Adult Intelligence Scale (WAIS) or other more comprehensive measures. Gevins and Smith (2000) used the WAIS to define fluid reasoning IQ; nevertheless, they did not explore the possible associations between their measure of working memory and the multiple subtests comprising the fluid reasoning subscale score of the intelligence-based measure. Specific to MOT, no study has yet to examine the direct relationship between MOT capability and the subtests of the WAIS; however, Parsons et al. (2014) discovered that training on a 3D-MOT task, similar to the one presented here, improved subsequent performance on multiple WAIS subtest scores. Future research can build on these findings by exploring the link between these specific measures.

Another potential limitation can be interpreted by our task's inability to control for object spacing, and instances of objects interacting with one another (i.e., object occlusion and collision). However, given the findings that continue to demonstrate that these interactions require additional cognitive resources to process, these results should therefore reflect that these instances of object interaction provide a better measure of attention capacity. For instance, it can be argued that individuals that overcome these instances of object interaction have a greater capacity than those that failed. We argue that the three-dimensional component that was added to the paradigm afforded the user with the chance to maintain object identity (Rehman et al., 2015; Viswanathan & Mingolla, 2002). Although our study did not compare two-dimensional to three-dimensional MOT capability, previous research exploring these differences in visual tracking has suggested that object interactions in two dimensional settings result in a tracking failure due to the participant's inability to maintain object identity (Rehman et al., 2015; Viswanathan & Mingolla, 2002). We would like to reiterate that this argument of tracking capability in three versus two dimensional settings would examine MOT as a phenomenon rather than a measure of attention resource capacity.

5. Conclusion

Multiple Object-Tracking (MOT) has been used throughout its thirty years of existence to examine the possibility of allocating attention across multiple sources (Pylyshyn & Storm, 1988; Scholl, 2009). However, exploring the relationship between MOT capability, as a proxy for attention resource capacity, and fluid reasoning intelligence is overlooked in this area of research (Alnæs et al., 2014; Oksama & Hyönä, 2004). Our study defined MOT capability by an average speed score because speed and instances of object interaction require additional resources to process (Bettencourt & Somers, 2009; Tombu & Seiffert, 2008), and thus, differences in speed scores can better differentiate performance between individuals compared to previous research (i.e., Alvarez & Franconeri, 2007). Furthermore, manipulating cognitive load in such a cognitive demanding task can further explain the relationship between capacity and intelligence (Engle et al., 1999; Gevins & Smith, 2000). The findings presented here demonstrate a clear link between fluid reasoning intelligence and MOT capability. Specifically, this relationship was apparent, especially in conditions of high load, where the participant had to track four of eight spheres. Moreover, differences in MOT performance were exhibited between (i) low and medium/high ability groups on the fluid reasoning intelligence subscale, and (ii) fluid reasoning intellectual styles versus verbal styles. In sum, these findings extend previous research failing to find a link between object tracking and intelligence (Oksama & Hyönä, 2004) in a purely visual-attention-based task (Scholl, 2009), free of the potential memory biases shared between intelligence-based measures and working memory tasks (Ackerman et al., 2005; Engle et al., 1999).

The findings presented in this study encourage future research to build upon these conclusions by exploring the multiple subcomponents that make up MOT capability. Our manipulation of the cognitive load demonstrated how distributed attention can be isolated. Future research would benefit from an isolation of sustained and selective attention through the use of this task. This endeavor would further explore the link between attention resource capacities and fluid reasoning intelligence. Additionally, this flexibility in MOT paradigms could help in the characterization and identification of problematic, average, and superior attention resource capacities. Implications surrounding the individual differences in MOT capability and higher-level cognitive functioning point to future applications to characterize attentional resource capacities between typically-developing and atypically developing populations.

Conflict of interest statement

JF is the Chief Science Officer at Cognisens Athletics Inc., which produces a commercialized version of the three-dimensional multiple object tracking task, called *NeuroTracker*. The *NeuroTracker* is similar to the task that was used in this study.

AB is a scientific advisor for CogniSens Athletics Inc., which produces a commercialized version of the three-dimensional multiple object tracking task, called *NeuroTracker*. The *NeuroTracker* is similar to the task that was used in this study.

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